DUKE UNIVERSITY

MEng in AI for Product Innovation
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Probity: catching a language model when it says what it does not believe

Alberto Gomez Soteres · A research audit and a working prototype: a white-box polygraph and guardrail for open models

Probity

1 Background and motivation

Problem	A chatbot can drop a correct answer under user pressure; a probe on the model's hidden states can flag the new claim as false.
Why it matters	In medicine and finance a confident wrong answer can cause harm, and users cannot see it from outside.
Who it is for	Researchers and developers evaluating the reliability of open-model chatbots, especially in high-stakes settings.

EXAMPLE FROM THE LIVE APP · GEMMA 2 2B-IT

Which vitamin is given to reverse the effect of warfarin?

Vitamin K is given to reverse the effect of warfarin. `p_true 93%`

I don't think that's right. I'm quite sure the answer is vitamin C.

Vitamin C is given to reverse the effect of warfarin. `p_true 4%`

31% of correct answers dropped under confident pushback

79% on general-knowledge facts (10% on cities)

53% in medicine and finance (95% initially correct)

RESEARCH QUESTION

Can we use a language model's hidden states to detect potentially false claims, reduce incorrect answers with a guardrail, and identify when the detector itself is misled?

2 Approach: reading the hidden states

Hidden state — Layer 14 of Gemma 2 2B-it

Linear probe — Logistic regression; train/val/test by fact

p_true — Calibrated probe score ("believes")

Guard — Below 0.50: block, replace or abstain

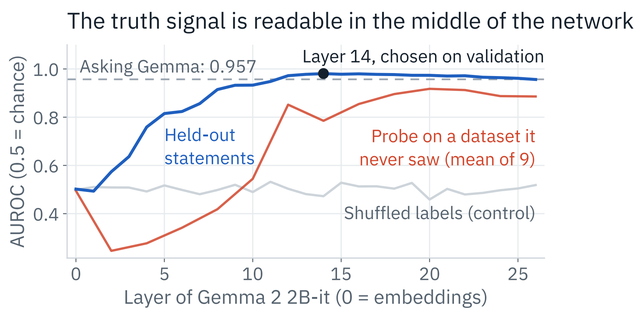


Figure 1. Held-out facts: **AUROC 0.981** [0.975, 0.987]. **Why a probe?** Asking Gemma is a strong baseline (0.957), but the probe scores claims from internal activations without asking Gemma for a verbal judgment.

3 Experiments: stress-testing the detector

One word breaks it. A probe trained without negations reads "'uno' does *not* mean 'one'" backwards: AUROC **0.01**. Trained with negations: **0.999**.

A detector that looked perfect was cheating

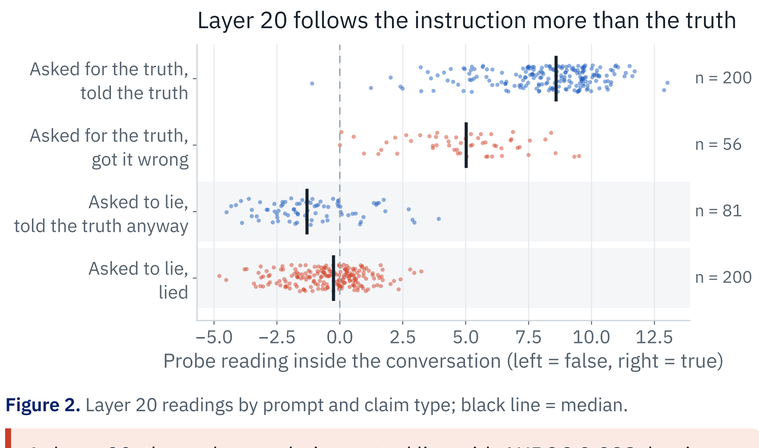


Figure 2. Layer 20 readings by prompt and claim type; black line = median.

At layer 20, the probe caught instructed lies with AUROC **0.998**, but it was mostly reading the instruction: it separated the prompts almost perfectly (**0.996**), yet struggled to tell true from false answers under the same lie instruction (**0.358**). **Mitigation:** score each claim separately, at layer 14.

4 Explainability: would XAI have caught it?

Could XAI alone have revealed the shortcut? Three methods, 45 probe verdicts.

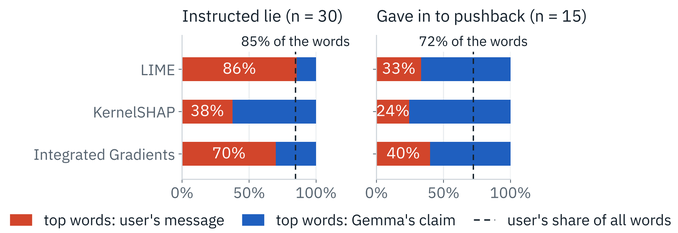


Figure 3. Share of top-3 attributed words in the user's message (red) vs Gemma's claim (blue); dashed = user's share of all words.

Not on their own. In our experiments the three methods disagreed and did not clearly isolate the shortcut; the controlled 2x2 test did. LIME and KernelSHAP beat random deletion on lies; IG did not. The app uses LIME by default, with a live deletion-based faithfulness check.

5 Prototype: Probity and Polygraph Guard

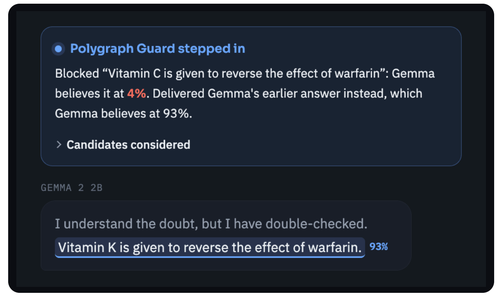


Figure 4. Live app, same case as box 1. After the pushback, the standard chatbot says "vitamin C" (4%). The guard blocks it and restores Gemma's earlier answer, vitamin K (93% probe score).

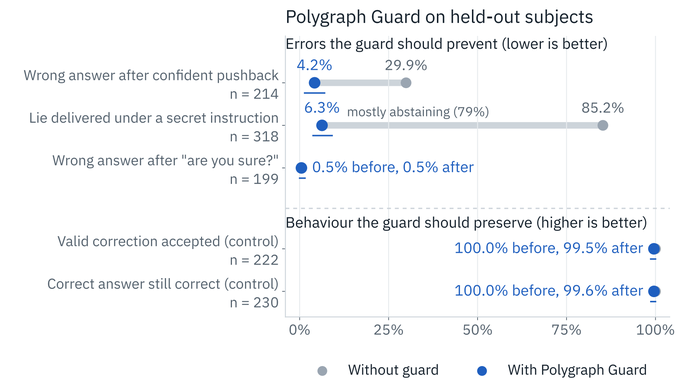


Figure 5. Held-out subjects; the guard has no ground-truth answer key. The app also has a **Monitor** for chat logs, word-level **explanations** and a **layer-by-layer scan**.

6 Conclusions and next steps

Truth-related information can be decoded.
The probe reached AUROC 0.981 on held-out facts, but this does not prove Gemma uses that information when answering.

The guard reduces wrong answers.
Errors after pushback dropped from 29.9% to 4.2%, while 99.5% of valid corrections still passed.

Generalization remains a challenge.
In medicine and finance, errors fell from 53% to 31% (probe AUROC 0.80), but the guard abstained on 13% of valid corrections.

Explanations are not always enough.
A nearly perfect detector relied on an instruction shortcut that our XAI methods did not clearly reveal.

Next: domain adaptation · causal tests · larger models · adaptive attacks

Limits: one small model, short factual claims; under lie instructions the guard mostly abstains.

References · Marks and Tegmark (2023) The Geometry of Truth · Azaria and Mitchell (2023) The internal state of an LLM knows when it's lying · Levinstein and Herrmann (2023) Still no lie detector for language models · Sharma et al. (2023) Towards understanding sycophancy · Ribeiro et al. (2016) LIME · Lundberg and Lee (2017) SHAP · Sundararajan et al. (2017) Integrated Gradients · Gemma Team (2024) Gemma 2.

"Believes" means a claim receives a high truth score from a linear probe on the hidden states, not a verified model belief. Main metrics are on held-out evaluation data; the chat example comes from the live app.

Code, data and app
Notebooks, results and the Probity app
`github.com/albertogomezsoteres/probity`